NILOUFAR BABA AHMADI

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EDUCATION

M.Sc. Intelligent Adaptive Systems, Universität Hamburg — Oct 2024 – Present

B.Sc. Computer Science, Minor in Management, University of Tehran — Sep 2019 – Feb 2024

RESEARCH INTERESTS

Natural Language Processing, Information Retrieval, Machine Learning, Deep Learning, Retrieval-Augmented Generation (RAG)

PUBLICATIONS

- **CollEX – A Multimodal Agentic RAG System Enabling Interactive Exploration of Scientific Collections**, *Co-author* View publication
  Co-authored a publication on CollEX, a multimodal agentic Retrieval-Augmented Generation system designed to support interactive exploration of large scientific collections through a chat interface. The system combines textual and visual modalities and was demonstrated on more than 64,000 records across 32 scientific collections.

RESEARCH EXPERIENCE

Research Assistant / Research Intern, University of Hamburg — Jul 2023 – Present
Under Prof. Chris Biemann, and Dr. Martin Semmann — *Hamburg, Germany*

- *Mar 2026:* In collaboration with Jan Strich. Contributed to POL-App, a medical imaging and Retrieval-Augmented Generation (RAG) system for diagnostic support and illness reporting. Improved chat streaming, multilingual behavior, and project documentation using `Django`, `Next.js`, and `shadcn/ui`.

- *Nov 2025 – Feb 2026:* In collaboration with Jan Strich, Dr. Amyn Vogel and Felizia von Schweinitz, developed a domain-specific RAG assistant for IMV Lab to support impact measurement and reporting based on Sustainable Development Goals (SDG) indicators. Built a source-grounded question-answering system over internal PDF documents and designed retrieval and prompting strategies in German.

- *Mar 2025 – Sep 2025:* In collaboration with Jan Strich. Developed a RAG system for the Universität Hamburg Tool Catalogue to support conversational tool discovery. Designed the data layer with `SQLAlchemy` and `Chroma`, integrated an LLM through OpenWebUI, and built a `Flask` application for retrieval and response generation.

- *Nov 2024 – Mar 2025:* In collaboration with Florian Schneider. Developed a RAG system with a software-engineering focus, contributing to LLM integration, typed function-calling design, and system robustness. This work resulted in a co-authored publication. View publication.

- *Jul 2023 – Feb 2024:* In collaboration with Hans Ole Hatzel. Investigated few-shot and zero-shot large language models for word-sense disambiguation using ontologies and developed a research prototype for ontology-based word-sense classification.

Visiting Scholar, KU Leuven — Jul 2022 – Sep 2022
Under Prof. Hugo Van Hamme — *Leuven, Belgium*
Studied probability calibration in speech recognition and explored an MLP-based approach to improve confidence estimation.

Bachelor's Thesis, University of Tehran — Apr 2023 – Jul 2023
Under Prof. Bagher BabaAli — *Tehran, Iran*
Applied machine learning methods to classify mental disorders from EEG data.

TEACHING EXPERIENCE

Teaching Assistant / Chief Teaching Assistant, University of Tehran — Sep 2021 – Mar 2024
Tehran, Iran

- ***Fundamentals of Computer Science and Programming**, Sep 2021 – Jan 2022:* Teaching Assistant under Prof. Mousavian; supported lectures, guided students, and assisted with grading.
- ***Fundamentals of Computer Science and Programming**, Jan 2022 – Jan 2023:* Chief Teaching Assistant under Dr. Ebrahimnejad; supported course instruction, coordinated with other teaching assistants, and assisted with grading.
- ***Advanced Information Retrieval**, Sep 2023 – Mar 2024:* Chief Teaching Assistant under Prof. Bagher BabaAli; supported teaching, coordinated assistants, assisted with grading, and helped design student projects.

SELECTED PROJECTS

- **Histopathology Image Classification**
  Developed a histopathology image classification system using deep learning and data augmentation techniques to support medical image analysis.
- **Sentiment Analysis**
  Applied natural language processing and machine learning methods to classify social media posts as positive or negative.
- **Grapevine Leaves Classification**
  Built a deep learning pipeline for classifying grapevine leaves using pre-trained models and image-based feature learning.
- **Parkinson's Disease Classification**
  Developed a machine learning model for Parkinson's disease classification based on voice-related features using KNN and PCA.

SELECTED COURSEWORK

Master's: Bio-Inspired Artificial Intelligence (1.7), Research Methods (1.7), Knowledge Processing (1.7)
Bachelor's: Database (1.0), Foundations of Operating Systems (1.0), Advanced Information Retrieval (1.0), Scientific Writing (1.0), Artificial Intelligence (1.3), Image Processing (1.3)

SELECTED ONLINE COURSES

- **Applied Data Science Capstone** — IBM Skills Network — Sep 2023
- **Advanced Learning Algorithms** — DeepLearning.AI, Stanford University — Nov 2022
- **Databases and SQL for Data Science with Python** — IBM Skills Network — Oct 2022
- **Supervised Machine Learning: Regression and Classification** — DeepLearning.AI, Stanford University Sep 2022

TECHNICAL SKILLS

Programming: Python, C/C++, JavaScript, R, SQL
Tools: LaTeX, Git, Docker, Flask, Django, Next.js, SQLAlchemy, Chroma, OpenWebUI
Languages: English (fluent), Persian (fluent), German (B1)